

OSSP

Question:

How to give read and write permissions using Unix commands.
A very simple example is to create a text file with a C program, then use chmod.

Solution:

```
ubuntu@ubuntu:~$ nano permissions.c

GNU nano 8.7.1 permissions.c
#include <stdio.h>

int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");

    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }

    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);

    printf("File created successfully.\n");
    return 0;
}

ubuntu@ubuntu:~$ gcc permissions.c -o permissions
ubuntu@ubuntu:~$ ./permissions
File created successfully.
ubuntu@ubuntu:~$

ubuntu@ubuntu:~$ ls -l sample.txt
-rw-rw-r-- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

Today's Tasks

1) Give Read and Write only to the owner

```
ubuntu@ubuntu:~$ chmod 600 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
-rw----- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

2) Give Read and Write to the owner and group

```
ubuntu@ubuntu:~$ chmod 660 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
-rw-rw---- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

3) Only owner can read and write

```
ubuntu@ubuntu:~$ chmod 600 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
-rw----- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

4) Owner read/write, everyone else read-only

```
ubuntu@ubuntu:~$ chmod 644 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
-rw-r--r-- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

5) Owner can do everything, others can only read and execute

```
ubuntu@ubuntu:~$ chmod 755 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
-rwxr-xr-x 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

6) Owner and group can read/write

```
ubuntu@ubuntu:~$ chmod 660 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
-rw-rw---- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

7) Only owner can execute

```
ubuntu@ubuntu:~$ chmod 100 sample.txt
ubuntu@ubuntu:~$ ls -l sample.txt
---x----- 1 ubuntu ubuntu 28 Aug 17 05:43 sample.txt
ubuntu@ubuntu:~$
```

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing

→ 600

9) Owner: read + write; Group: read; Others: read →

644

10) Owner: read + write + execute; Group: read + execute;

Others: read + execute →

755

11) Owner: read + write; Group: read + write; Others:

nothing →

660

12) Everyone: read only →

444

